Python Activity-Based Learning Lab Outputs

Execution Results & Terminal Screenshots • Cases 01 to 10

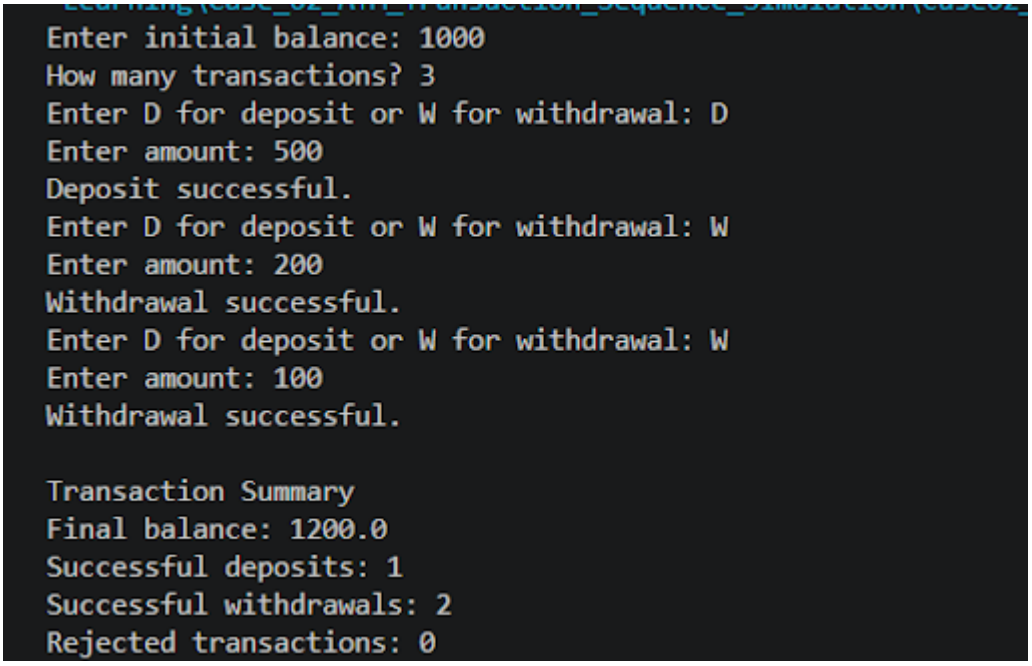
Case 01: Attendance Recovery Computation

`case01_attendance_recovery.py`



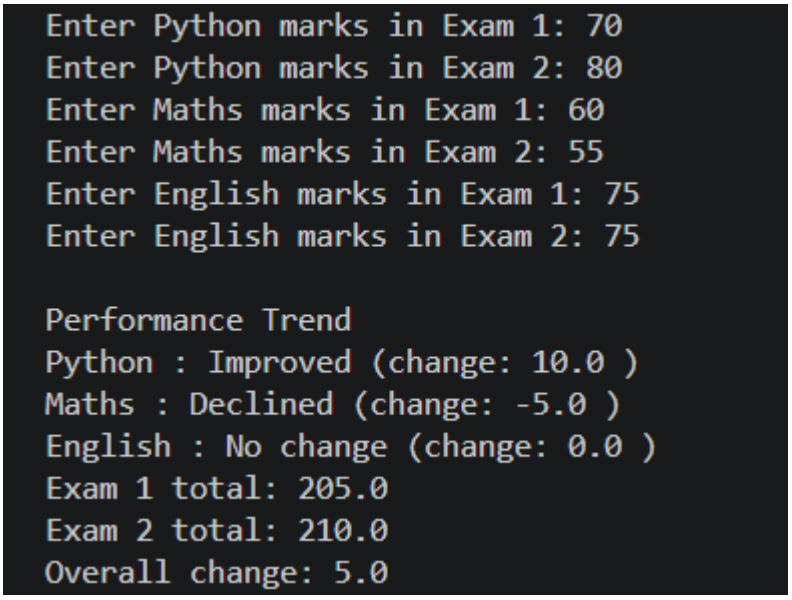
Case 02: ATM Transaction Sequence Simulation

`case02_atm_transaction.py`



Case 03: Student Performance Trend Tracker

`case03_performance_trend.py`



Case 04: Password Policy Validation Engine

case04_password_policy.py

```
Enter a sample password to check: Abc@1234
Password meets all sample rules.
PS C:\Users\shabnum baig\Downloads\Python-Activity-Based-Lear
```

Case 05: MCQ Scoring with Negative Marking

case05_mcq_scoring.py

```
Enter 5 answers separated by spaces (use - for blank): A B C D A
Correct answers: 3
Incorrect answers: 2
Unanswered questions: 0
Final score: 2.5 out of 5
```

Case 06: Class Performance Analytics & Top Performer

case06_marks_matrix.py

```
Asha total: 253 average: 84.33
Ravi total: 240 average: 80.0
Sara total: 272 average: 90.67

Subject averages:
Python : 83.0
Maths : 85.67
English : 86.33
Top performer: Sara with 272 marks
```

Case 07: Product Catalogue Audit & Low Stock Warning

`case07_catalogue_audit.py`

```
Product Catalogue Audit
P101 inventory value: 500.0
P102 inventory value: 360.0
  Replenishment needed; current stock: 3
P103 inventory value: 0.0
  Replenishment needed; current stock: 0
P104 inventory value: 320.0
Total inventory value: 1180.0
```

Case 08: Mess Food Wastage Threshold Alert System

`case08_food_wastage.py`

```
Monday leftovers: 8 wastage: 8.0 %
Tuesday leftovers: 20 wastage: 16.67 %
  Warning: Wastage exceeded 10 %
Wednesday leftovers: 5 wastage: 5.56 %
Thursday leftovers: 20 wastage: 18.18 %
  Warning: Wastage exceeded 10 %
Friday leftovers: 30 wastage: 30.0 %
  Warning: Wastage exceeded 10 %
```

Case 09: Canteen Menu Order & Inventory Stock Management

`case09_canteen_stock.py`

```
Canteen Menu
1 Sandwich - Rs. 50 | Stock: 10
2 Samosa - Rs. 15 | Stock: 20
3 Juice - Rs. 25 | Stock: 12
Enter number of orders: 3
enter item number(1-3):1
Enter quantity: 3
Order accepted. Bill: Rs. 150
enter item number(1-3):2
Enter quantity: 3
Order accepted. Bill: Rs. 45
enter item number(1-3):3
Enter quantity: 1
Order accepted. Bill: Rs. 25

Daily Sales: Rs. 220
Remaining Stock:
Sandwich : 7
Samosa : 17
Juice : 11
```

Case 10: Dynamic Hollow Diamond Pattern Generator

`case10_diamond_pattern.py`

```
Enter diamond size (2-20): 5
    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *
```